

Performance Testing

Date	06 July 2026
Team ID	
Project Name	Smart Lender - Loan Eligibility Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache Jmeter [e.g., JMeter, Locust, k6, Postman]
Type of Testing	Load Testing [e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	Loan prediction API [Specify which feature or endpoint was tested]
Test Environment	Local (Flask Application) [e.g., Local / Cloud / Hosted Server]
Test Date	06 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user prediction	1	30	Response within 2 sec
2	Multiple users	20	60	All requests processed successfully
3	Peak loan test	50	120	Response time below 5 sec
4	Continuous load	100	180	No server crash, low error rate

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Good performance
2	Response Time (Max)	< 5 seconds	3.4 sec	Pass	Within limit
3	Throughput (Req/sec)		42 req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	52%	Pass	Normal usage
6	Memory Utilization	< 80%	60%	Pass	Stable memory usage

Step 4: Observations & Analysis

- 1.The Smart Lender application handled concurrent users efficiently.
- 2.Average response time remained below 2 seconds.
- 3.No request failures were observed during testing.

Bottleneck Identified:
Slight increase in response time under heavy load 100 users, but still within acceptable limits.

Optimization Steps Taken:
 1.Optimized flask API response handling.
 2.Reduced unnecessary processing.
 3.Improved model loading efficiency using joblib.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1 : Apache Jmeter Test Plan and Summary Report.

Screenshot 2 : Jmeter Graphs Results and Aggregate Report.

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]